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# Per- and Polyfluoroalkyl Substance Levels in Commercial and Home-Produced Eggs in Croatia

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## Abstract

The aim of this study was to measure per- and polyfluoroalkyl substance (PFAS) levels in eggs collected in Croatia and to identify differences between commercially produced eggs (cage, barn, and organic) and home-produced eggs (HPE). Thirty PFAS compounds were analyzed using high-performance liquid chromatography coupled with high-resolution mass spectrometry. In HPE, the highest detection frequencies above the limit of quantification were observed for perfluorooctane sulfonic acid (PFOS) at 67.6%, perfluorononanoic acid (PFNA) at 43.2%, perfluoro-n-decanoic acid (PFDA) at 43.2%, and perfluoro-n-dodecanoic acid (PFDoDA) at 35.8%. Perfluorooctanoic acid (PFOA) was detected only in HPE. Furthermore, HPE exhibited significantly higher mean lower bound (LB) and upper bound (UB) levels for all measured compounds, as well as for the sum of the four main PFAS (∑4PFAS: PFOS, PFOA, PFNA, and perfluorohexane sulfonic acid [PFHxS]), with values of 0.263 and 0.44 µg/kg, respectively. Cage eggs showed the lowest LB and UB levels. The dietary contribution of ∑4PFAS to the established tolerable weekly intake (TWI) limit of 4.4 ng/kg bw indicated that children up to nine years old are the most vulnerable to exposure, particularly infants and toddlers consuming HPE. Significantly lower exposure was observed with cage eggs; therefore, consumers are advised to prefer these eggs.

**Keywords:** PFAS; eggs; food contaminants; risk assessment; food safety; Croatia

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## 1. Introduction

Contamination with per- and polyfluoroalkyl substances (PFAS) is a global issue today, affecting terrestrial and aquatic ecosystems as well as food sources. These compounds, due to their unique chemical properties—such as high surface activity and both hydrophobic and lipophobic characteristics—have been used for decades in numerous consumer products for a wide range of applications, as well as in industrial processes [1,2]. The chemical structure of PFAS, particularly the strong carbon–fluorine bond, renders them exceptionally resistant to environmental and biological degradation [3]. Because of these persistent properties, PFAS readily migrate through environmental compartments such as soil and

water and bioaccumulate in various environmental and food matrices [4,5]. Research has shown that PFAS compounds perfluorooctanoic acid (PFOA) and perfluorooctane sulfonic acid (PFOS) are highly persistent in the environment and bioaccumulate in organisms due to their protein-binding affinity [2,6,7]. These two long-chain PFAS compounds, along with perfluorohexane sulfonic acid (PFHxS), have been listed as Persistent Organic Pollutants (POPs) under the Stockholm Convention because of their persistence, bioaccumulation, and significant adverse effects on human health and the environment [8]. Additionally, long-chain perfluorocarboxylic acids (LC-PFCA), their salts, and related compounds are candidates for inclusion as POPs due to their capacity for long-range transport and their numerous detrimental impacts on human health and the environment [9].

The primary source of PFAS exposure for the general population in Europe is food, accounting for more than 70% of total intake [2,10]. While food contributes the most to overall PFAS exposure, drinking water, household dust, and contact with common objects also play a role [11,12]. By analyzing a large dataset of concentrations of 17 PFAS compounds in food collected from EU member states, European Food Safety Authority (EFSA) determined that, in addition to the food groups "Fish meat" and "Fruit and fruit products, "Eggs and egg products" are also major sources of human exposure to PFAS compounds [10]. In adults, four compounds—PFOS, PFOA, perfluorononanoic acid (PFNA), and PFHxS—account for approximately 46% of the total sum of all PFAS compounds for which exposure was calculated, with PFOS alone contributing 30%. To protect consumer health, the European Commission adopted Regulation 2022/2388, which sets maximum levels (ML) for four individual PFAS compounds (PFOA, PFOS, PFNA, and PFHxS) as well as their total amount in eggs and other foods of animal origin, including muscles, meat, offal, and crustaceans [13]. These ML are specified in Commission Regulation (EU) 2023/915, which replaced and repealed Regulation 1886/2006 and came into forse in April 2023 [14].

PFAS compounds have multiple health effects, as identified and explained by various scientific studies. The results of epidemiological studies show that long-term exposure to PFAS compounds increases serum cholesterol levels, which is a risk factor for the development of cardiovascular diseases [10]. A link has also been established between PFAS exposure and increased serum levels of the enzyme alanine transaminase (ALT), a marker of liver damage [10]. Other negative health impacts include changes in immune and thyroid function, disruptions in lipid and insulin regulation [15,16]. PFASs have also been associated with elevated systolic and diastolic blood pressure and hypertension in highly exposed young adults [17]. These compounds adversely affect conception, pregnancy, and child development [15]. The two best-studied PFAS compounds, PFOA and PFOS, have been linked to reduced kidney function [18], as well as non-Hodgkin lymphoma, testicular, kidney, prostate, and ovarian cancers [19]. The International Agency for Research on Cancer (IARC) has classified PFOA, one of the most commonly used PFAS compounds, as Group 2B, indicating it is possibly carcinogenic to humans [20].

Based on available data, EFSA has established tolerable weekly intake (TWI) limits for PFOS and PFOA at 13 and 6 ng/kg body weight (bw) per week, respectively [21]. Subsequently, EFSA updated its scientific opinion, establishing a cumulative daily reference point intake value of 0.63 ng/kg bw per day for the four main PFAS: PFOA, PFNA, PFHxS, and PFOS [10]. This reference value, derived using a physiologically based pharmacokinetic model, represents the maternal daily intake expected to result in a serum concentration of 17.5 ng/mL in children. This concentration corresponds to the benchmark dose identified by EFSA for reduced antibody responses to childhood vaccinations, which was considered the most sensitive and health-relevant critical effect. However, given the long half-life of these PFAS, the risk is determined by the cumulative body burden over time, so that a TWI

of 4.4 ng/kg bw was considered more appropiate. In the same year, the Agency for Toxic Substances and Disease Registry (ATSDR) and the Environmental Protection Agency (EPA) of the United States revised the estimated minimal risk levels (MRLs) for daily human exposure to PFOA and PFOS to 3.0 and 2.0 ng/kg bw per day, respectively [22].

Public perception of food safety today often favors organic food or free-range animal farming as healthier alternatives to conventional food or large-scale controlled farming systems. However, recent research indicates that such foods are highly exposed to environmental persistent organic pollutants (POPs), including PFAS [23–25]. Additionally, research indicates that bioaccumulation of PFAS depends on the length of the carbon chain, with longer chains accumulating more persistently than shorter ones [2,26]. Furthermore, the concentrations of these compounds in eggs or food depend on their proximity to anthropogenic sources of pollution [4,27].

Eggs are an important source of nutrients such as proteins, fats, vitamins, and trace elements in the diet of the Croatian population and in other countries worldwide. They have a well-established positive effect on bodily functions [28]. Total egg production in Croatia reached 39,000 tons in 2023 [29], equivalent to approximately 666 million pieces [30]. In 2024, a 3.6% increase in chicken egg production was recorded. According to statistical data, the average Croatian consumes 176 eggs per capita (10.6 kg), compared to 218 eggs per capita (13.1 kg) in European Union member states [28]. According to the latest data from the competent authority in Croatia, the number of poultry farms has increased over the past five years. Of these farms, 18.8% use cage systems, 21.6% use floor systems (barn), 56.1% use home (domestic) systems, and 8% employ organic production systems [31]. However, the majority of laying hens—67.3%—are kept in cages, followed by 29.9% in floor systems (barn). Home-produced laying hens account for only 2.6%, and just 0.2% are raised under organic production systems [28].

The purpose of this study was to investigate the presence of PFAS in eggs. This is the first study to assess the concentrations of these compounds in Croatia. Differences in PFAS levels between commercially produced eggs (caged, barn—floor systems and organic) and home-produced eggs were examined. The results were compared with previous studies that reported significant differences in PFAS concentrations across various poultry farming methods [23,32]. Additionally, estimates of dietary intake of these compounds through egg consumption were calculated for different age groups within the Croatian population and compared with the established TWI values.

## 2. Materials and Methods

### *2.1. Sample Collection*

A total of 424 egg samples were collected during 2024 and 2025 from various egg producers across Croatia. Some samples were purchased from different supermarkets, while others were collected by the State Veterinary Inspectorate as part of national monitoring programs for food of animal origin. The remaining samples were obtained from small free-range egg production farms or were homegrown eggs from backyard chickens. According to the type of farming for laying hens used in egg production, eggs were divided into the following groups: cage ($n$ = 165), barn ($n$ = 91), organic ($n$ = 20) and home-produced ($n$ = 148).

After collection, the egg samples were homogenized in the laboratory, labeled, and stored at −18 °C until analysis.

### 2.2. Standards and Chemicals

All analytical standards solutions, purchased from various suppliers, were with a purity > 98%. Perfluorobutanoic acid (PFBA), perfluoro-n-pentanoic acid (PFPeA), perfluorohexanoic acid (PFHxA), perfluoroheptanoic acid (PFHpA), perfluorononanoic acid (PFNA), perfluoro-n-decanoic acid (PFDA), perfluoro-n-undecanoic acid (PFUnDA), perfluoro-n-dodecanoic acid (PFDoDA), perfluorotridecanoic acid (PFTrDA), perfluorotetradecanoic acid (PFTeDA), perfluorobutanesulfonic acid (PFBS), perfluoropentanesulfonic acid (PFPeS), perfluorohexanesulfonic acid (PFHxS), perfluoroheptanesulfonic acid (PFHpS), perfluorodecanesulfonic acid (PFDS), 3-H-Perfluoro-4,8-dioxanonanoic acid (DONA), perfluoro-2-propoxypropanoic acid (HFPO-DA, GenX), 1H,1H,2H,2H-Perfluorohexane sulfonic acid (4:2 FTS), 1H,1H,2H,2H-Perfluorooctane sulfonic acid (6:2 FTS), and 1H,1H,2H,2H-Perfluorodecane sulfonic acid (8:2 FTS) were purchased from Dr. Ehrenstorfer LGC Group (Teddington, UK). Pentadecafluorooctanoic acid (PFOA) was obtained from Supelco, Merck KGaA (Darmstadt, Germany). Perfluorononanesulfonic acid sodium salt (PFNS) was purchased from HPC Standards GmbH (Cunnersdorf, Germany). Sodium perfluoro-1-undecanesulfonate (PFUnDS), sodium perfluoro-1-dodecanesulfonate (PFDoDS), sodium perfluoro-1-tridecanesulfonate (PFTrDS), potassium 9-chlorohexadecafluoro-3-oxanonane-1-sulfonate (9Cl-PF3ONS), and potassium 11-chloroeicosafluoro-3-oxaundecane-1-sulfonate (11Cl-PF3OUdS) were obtained from Wellington Laboratories Inc. (Guelph, ON, Canada). Perfluorooctanesulphonamide (FOSA), Capstone A, and Capstone B were obtained from CPAchem (Stara Zagora, Bulgaria). A mixture of perfluorooctanesulfonic acid (PFOS) isomers contained 79.3% linear PFOS (L-PFOS) and 20.7% branched PFOS (br-PFOS) was obtained from Cambridge Isotope Laboratories (Andover, MA, USA). The isotopically labeled standards M3HFPO-DA and MPFAC-24ES solution were purchased from Wellington Laboratories Inc. (Guelph, ON, Canada).

Acetonitrile (ACN) and methanol (MeOH) (ULC/MS-CC/SFC grade, ≥99.9%) were purchased from Biosolve Chimie (Dieuze, France). Ammonium acetate (LC–MS grade, ≥99%) and magnesium sulfate anhydrous ($MgSO_4$) (reagent grade, ≥97%) were obtained from Sigma-Aldrich (Steinheim, Germany), while sodium chloride (NaCl) (analytical grade, ≥99.5%) was purchased from Supelco, Merck KGaA (Darmstadt, Germany). SPE adsorbents Chromabond C18 (bulk) and Bond Elut Carbon S (bulk) were purchased from Macherey-Nagel GmbH & Co. KG (Düren, Germany) and Agilent Technologies (Santa Clara, CA, USA), respectively. Ultrapure water ($H_2O$) was obtained using a Mili-Q system (Millipore®, Bedford, MA, USA).

Standard mixtures of 30 PFAS analytes were prepared in methanol at concentrations of 1000, 100, 10, and 1 ng/mL. A mixture of isotopically labelled standards (IS) was prepared in methanol at a concentration of 50 ng/mL. All solutions were stored at −20 to −10 °C.

### 2.3. Sample Preparation

Egg samples were homogenized using a knife mill Grindomix GM200 (Retsch GmbH, Haan, Germany). Sample preparation was performed using a slightly modified Quick, Easy, Cheap, Effective, Rugged, and Safe (QuEChERS) method as described by the European Union Reference Laboratory for halogenated POPs in Feed and Food (EURL POPs, 2024). A homogenized sample (2 g) was weighed into a 50 mL polypropylene (PP) centrifuge tube and spiked with 20 µL of IS at 50 ng/mL. The samples were shaken for 1 min at 2500 rpm using a multi-tube vortexer (VWR International GmbH, Ulm, Germany) and then left to stand in the dark for 15 min.

Extraction was carried out using 5 mL of acetonitrile. The samples were shaken for 1 min at 2500 rpm and sonicated for 15 min in an ultrasonic bath Grant (Grant instruments, Cambridge, UK), followed by centrifugation at 4000 rpm for 10 min at 20 °C using a

Rotanta 460R centrifuge (Hettich Zentrifugen, Tuttlingen, Germany). The supernatant was transferred to a clean 50 mL PP tube. The extraction step was repeated once again with 5 mL of acetonitrile using the same procedure as previously described.

The two supernatants were combined, after which a salt mixture containing 2.0 g $MgSO_4$, 0.5 g NaCl, 0.1 g C18, and 0.1 g Carbon S was added and promptly shaken by hand for 10 s. Shaking was continued on a Multi Reax vortex (Heidolph Scientific Products GmbH, Schwabach, Germany) for 1 min. The samples were then centrifuged at 4000 rpm for 15 min at 20 °C. The supernatant was transferred to a 15 mL PP tube, evaporated to dryness under a gentle nitrogen stream at 40 °C using a MultiVap 54 automatic evaporation system (LabTech Srl, Sorisole (BG), Italy) and reconstituted with 500 µL of mobile phases A and B (1:1, *v*/*v*). Afterwards, a mixture was vortexed for 10 s using an IKA® Vortex MS2 Minishaker (IKA®-Werke GmbH & Co. KG, Staufen, Germany), followed by sonication for 5 min, vortexed again for 10 s, and centrifuged in 2 mL Eppendorf Safe-Lock tubes (Eppendorf, Hamburg, Germany) at 13,000 rpm for 15 min at 4 °C using an ultracentrifuge Thermo Scientific SL16R (Thermo Fisher Scientific, Waltham, MA, USA). Finally, the supernatant was transferred into a PP vial prior to instrumental analysis.

### *2.4. Quantitative Analysis*

The analysis was performed using a UHPLC 1290 Infinity II equipped with an InfinityLab PFC-free HPLC Conversion Kit and coupled to a Triple Quad LC/MS 6495C mass spectrometer with a Jet Stream Technology Ion Source (AJS ESI). A Zorbax RRHD Eclipse Plus C18 column (2.1 × 100 mm, 1.8 µm) with a Zorbax RRHD Eclipse Plus C18 guard column (2.1 × 5 mm, 1.8 µm) was used for chromatographic separation. The column temperature was set to 25 °C. Additionally, to prevent co-elution of interferences, the InfinityLab PFC Delay column (4.6 × 30 mm, 1200 bar) was installed between the solvent pump and the injector. All components of the UHPLC-MS/MS system were acquired from Agilent Technologies (Santa Clara, CA, USA). The mobile phases were 2 mM ammonium acetate in $H_2O$ (A) and 2 mM ammonium acetate in MeOH (B). A gradient method with a flow rate of 0.3 mL/min was used. The gradient started at 25% B, increased to 60% B at 2.5 min, 80% B at 10 min, 90% B at 12 min (held until 14.75 min), and 100% B at 15 min (held until 16 min), before returning to the initial conditions until 21.5 min. The injection volume was 5 µL.

The mass spectrometer was operated with electrospray ionization (ESI) in negative polarization mode using dynamic multiple reaction monitoring (dMRM). dMRM transitions for analytes and IS are summarized in Table 1. Source parameters were as follows: gas temperature 250 °C, gas flow 11 L/min, nebulizer 25 psi, sheath gas temperature 375 °C, sheath gas flow 11 L/min, capillary voltage 2500 V, nozzle voltage 0 V, and EMV 250 V. Ion Funnel parameters were 90 V (high pressure RF) and 60 V (low pressure RF). The collision cell accelerator voltage (CAV) was set to 5 V for all analytes. The fragmentor voltage was set to a fixed value of 166 V, which could not be adjusted.

Data processing was performed using MassHunter Workstation software (Version 10.1). Quantification was performed using a seven-point linear calibration curve with 1/x weighting, ranging from 0.1 to 20 ng/mL for analytes and 2 ng/mL for all internal standards. Samples exceeding the highest calibration point were reanalyzed using an expanded nine-point calibration curve, which included two additional points at 50 and 200 ng/mL.

**Table 1.** Dynamic multiple reaction monitoring transitions for target PFAS compounds and isotopically labelled standards (IS).

| Analyte | IS | Precursor Ion (*m*/*z*) | Product Ion (*m*/*z*) | Collision Energy (eV) | Retention Time (Min) |
|---|---|---|---|---|---|
| Perfluoroalkyl carboxylic acids (PFCA) | | | | | |
| PFBA | $^{13}C_4$-PFBA | 213 | 169 | 6 | 3.49 |
| PFPeA | $^{13}C_5$-PFPeA | 263 | 219 | 4 | 4.86 |
| PFHxA | $^{13}C_5$-PFHxA | 313 | **269**<br>119 | 6<br>22 | 5.72 |
| PFHpA | $^{13}C_4$-PFHpA | 363 | **319**<br>169 | 6<br>18 | 6.77 |
| PFOA | $^{13}C_8$-PFOA | 413 | **369**<br>169 | 6<br>18 | 7.99 |
| PFNA | $^{13}C_9$-PFNA | 463 | **419**<br>219 | 10<br>18 | 9.28 |
| PFDA | $^{13}C_6$-PFDA | 513 | **469**<br>269 | 6<br>18 | 10.53 |
| PFUnDA | $^{13}C_7$-PFUnDA | 563 | **519**<br>269 | 12<br>16 | 11.68 |
| PFDoDA | $^{13}C_2$-PFDoDA | 613 | **569**<br>319 | 12<br>22 | 12.69 |
| PFTrDA | $^{13}C_2$-PFDoDA | 663 | **619**<br>169 | 14<br>34 | 13.42 |
| PFTeDA | $^{13}C_2$-PFTeDA | 713 | 669<br>**169** | 12<br>32 | 13.92 |
| Perfluoroalkyl sulfonic acids (PFSA) | | | | | |
| PFBS | $^{13}C_3$-PFBS | 299 | 99<br>**80** | 44<br>36 | 5.01 |
| PFPeS | $^{13}C_5$-PFHxA | 349 | 99<br>**80** | 36<br>40 | 5.82 |
| PFHxS | $^{13}C_3$-PFHxS | 399 | 119<br>99<br>**80** | 36<br>40<br>48 | 6.82 |
| PFHpS | $^{13}C_8$-PFOA | 449 | 99<br>**80** | 46<br>50 | 8.01 |
| L-PFOS | $^{13}C_8$-PFOS | 499 | 99<br>**80** | 56<br>56 | 9.26 |
| br-PFOS | $^{13}C_8$-PFOS | 499 | 169<br>99<br>**80** | 56<br>56<br>56 | 8.7–9.1<br>(2 peaks) |
| PFNS | $^{13}C_2$-8:2 FTS | 549 | 99<br>**80** | 48<br>76 | 10.49 |
| PFDS | $^{13}C_7$-PFUnDA | 599 | 99<br>**80** | 60<br>60 | 11.61 |
| PFUnDS | $^{13}C_2$-PFDoDA | 649 | **99**<br>80 | 55<br>43 | 12.61 |
| PFDoDS | $^{13}C_2$-PFTeDA | 699 | 99<br>**80** | 60<br>64 | 13.34 |
| PFTrDS | $^{13}C_2$-PFTeDA | 749 | **99**<br>80 | 59<br>35 | 13.87 |

**Table 1.** *Cont.*

| Analyte | IS | Precursor Ion (*m*/*z*) | Product Ion (*m*/*z*) | Collision Energy (eV) | Retention Time (Min) |
|---|---|---|---|---|---|
| Perfluoroalkane sulfonamides | | | | | |
| FOSA | $^{13}C_8$-FOSA | 498 | 478<br>**78** | 21<br>38 | 11.82 |
| DONA | $^{13}C_3$-PFHxS | 377 | **251**<br>85 | 12<br>36 | 6.87 |
| HFPO-DA (GenX) | M3HFPO-DA ($^{13}C_3$-HFPO-DA) | 285 | 185<br>**169**<br>119 | 16<br>4<br>30 | 6.00 |
| 9Cl-PF3ONS | $^{13}C_2$-8:2 FTS | 531 | **351**<br>99<br>83 | 28<br>28<br>32 | 10.00 |
| 11Cl-PF3OUdS | $^{13}C_8$-FOSA | 631 | **451**<br>83 | 32<br>32 | 12.21 |
| Capstone A | $^{13}C_9$-PFNA | 527 | **507**<br>181 | 7<br>11 | 9.71 |
| Capstone B | $^{13}C_8$-PFOS | 569 | **549**<br>223 | 11<br>13 | 9.20 |
| Fluorotelomer sulfonates (FTS) | | | | | |
| 4:2 FTS | $^{13}C_2$-4:2 FTS | 327 | **307**<br>81<br>80 | 16<br>44<br>32 | 5.63 |
| 6:2 FTS | $^{13}C_2$-6:2 FTS | 427 | **407**<br>81<br>80 | 28<br>44<br>44 | 7.91 |
| 8:2 FTS | $^{13}C_2$-8:2 FTS | 527 | **507**<br>80 | 32<br>52 | 10.50 |
| Isotopically labelled standards (IS) | | | | | |
| M3HFPO-DA ($^{13}C_3$-HFPO-DA) | - | 287 | 185 | 20 | 6.00 |
| MPFAC-24ES | | | | | |
| $^{13}C_4$-PFBA | - | 217 | 172 | 5 | 3.49 |
| $^{13}C_5$-PFPeA | - | 268 | 223 | 5 | 4.86 |
| $^{13}C_5$-PFHxA | - | 318 | 273 | 10 | 5.72 |
| $^{13}C_4$-PFHpA | - | 367 | 322 | 7 | 6.77 |
| $^{13}C_8$-PFOA | - | 421 | 376 | 7 | 7.99 |
| $^{13}C_9$-PFNA | - | 472 | 427 | 7 | 9.28 |
| $^{13}C_6$-PFDA | - | 519 | 474 | 10 | 10.53 |
| $^{13}C_7$-PFUnDA | - | 570 | 525 | 10 | 11.68 |
| $^{13}C_2$-PFDoDA | - | 615 | 570 | 9 | 12.69 |
| $^{13}C_2$-PFTeDA | - | 715 | 670 | 11 | 13.93 |
| $^{13}C_8$-FOSA | - | 506 | 78 | 39 | 11.82 |
| $^{13}C_3$-PFBS | - | 302 | 80 | 38 | 5.00 |
| $^{13}C_3$-PFHxS | - | 402 | 99 | 39 | 6.82 |
| $^{13}C_8$-PFOS | - | 507 | 99 | 47 | 9.26 |
| $^{13}C_2$-4:2 FTS | - | 329 | 309 | 24 | 5.63 |
| $^{13}C_2$-6:2 FTS | - | 429 | 409 | 28 | 7.91 |
| $^{13}C_2$-8:2 FTS | - | 529 | 509 | 28 | 10.50 |

The product ion (in bold) was used for compound quantitation.

### 2.5. Validation Data

The analytical parameters, including specificity, selectivity, linearity, trueness, precision, limit of quantification (LOQ), and measurement uncertainty were evaluated in accordance with Commission Implementing Regulation (EU) 2022/1428 and the EURL Guidance Document [33,34]. According to Commission Recommendation (EU) 2022/1431, the LOQ for PFOS, PFOA, PFNA, and PFHxS in eggs should be $\leq$0.30 µg/kg [35].

Specificity and selectivity were tested by analyzing 20 blank samples. Linearity was assessed using linear regression across seven concentration levels ranging from 0.1 to 20 ng/mL. Linearity was confirmed for all analytes, with coefficients of determination ($R^2$) $\geq$ 0.98, using a 1/x weighting factor. Trueness, precision, and LOQ were determined using blank egg samples fortified at seven concentration levels (0.025, 0.05, 0.1, 0.15, 0.2, 0.5, and 2.5 µg/kg) in duplicate across six batches.

### 2.6. Quality Control and Quality Assurance

Each batch included a solvent calibration curve (0.1–20 ng/mL), a solvent blank, a blank egg sample, and a blank egg sample spiked at 0.1 and 0.2 µg/kg. Individual PFAS were quantified using their corresponding IS with the exception of 11Cl-PF3OUdS, 9Cl-PF3ONS, Capstone A, Capstone B, DONA, PFDoDS, PFDS, PFHpS, PFNS, PFPeS, PFTrDA, PFTrDS, and PFUnDS for which no specific IS were available. These analytes were quantified using the IS selected based on the closest retention time or similarity in functional group and size. Performance of the method was confirmed by successful participation in a proficiency testing (PT) organized by FAPAS (Fera Science LTD., Sand Hutton, York, UK).

### 2.7. Statistical Analysis

For the statistical analysis of descriptive parameters and differences between groups of results, Stata 13.1 for Windows (64-bit x86-64) (StataCorp LP, College Station, TX, USA) was used. Concentrations of each PFAS compound and the sum of the four main ∑4PFAS were expressed as mean, median, and standard deviation (SD).

The concentration values are expressed as lower bound (LB) and upper bound (UB) values. For PFAS analytes with concentrations below the limit of detection (LOD) and LOQ zero values are assigned for the LB, while the LOD (0.10 µg/kg) or LOQ (0.25 µg/kg) values are assigned for the UB.

The distribution of the data was assessed using the Shapiro–Wilk W test. Since the data were found to be non-normally distributed, statistical analyses comparing PFAS results among three types of farming laying hens were conducted using the Wilcoxon rank-sum test (Mann–Whitney U test). The significance level was set at $p < 0.05$.

### 2.8. Weekly Intake Estimation and Risk Characterization

Dietary intake of PFAS from egg consumption was estimated as weekly intake (WI, µg/kg bw/week) for the sum of four main PFAS compounds (∑4PFAS; PFOS + PFOA + PFNA + PFHxS) using the following equation:

$$WI = C \times MS \times 7 \quad (1)$$

where C is the mean concentration of ∑PFAS (µg/kg), and MS is the meal size (grams per egg portion per day) relative to body weight (g/kg bw/day). The LB scenario was applied for the calculation, meaning the zero value was used to account for the contribution of undetected analytes. The UB scenario was also applied for the calculation, meaning the LOQ value was used to account for the contribution of undetected analytes.

Egg consumption data for different age groups of both females and males in the Croatian population are available from the EFSA Comprehensive European Food Consumption

Database [36]. Data on consumption refer to the EFSA category "eggs and egg products" and to "consumers only". Average chronic egg consumption (g/kg bw/day) between both sexes, expressed as the mean for various age groups, was used as follows: infants (<1 year) 1.82, toddlers (1 to 3 years) 1.66, other children (3 to 9 years) 1.02, adolescents (10 to 17 years) 0.55, adults (18 to 65 years) 0.48, eldery (>65 years) 0.43.

To assess the potential risk posed by the consumption of eggs, the WI values were compared with the TWI limit of 4.4 ng/kg bw per week, established by EFSA for the sum of four main PFAS compounds (∑4PFAS). This comparison was expressed as a percentage of the TWI (%TWI) [10], using the following equation:

$$\% \text{ TWI} = \text{WI}/\text{TWI} \quad (2)$$

## 3. Results and Discussion

### 3.1. Validation Parameters

Validation parameters for PFAS compounds are presented in Table 2. Trueness, expressed as apparent recovery, ranged from 104.0–111.7% for PFOS, PFOA, PFHxS, and PFNA (compliance testing) and from 72.0–128.0% for other analytes (monitoring), while precision ranged from 5.6–17.4% and 7.9–24.8%, respectively. Both trueness (80–120% for compliance testing and 65–135% for monitoring) and precision ($\leq$20% and $\leq$25%, respectively) met the required performance criteria. Two analytes (PFBA and PFPeA) did not produce a second product ion and therefore did not meet the minimum product ion criterion. The LOQ was defined as the lowest fortification level meeting identification (two product ions, signal-to-noise (S/N) ratio $\geq$ 3, ion ratio within $\pm$30% of calibration standards, relative retention time $\leq$ 1% for analytes with isotopically labelled analogues, and complete peaks overlap) and performance criteria (trueness and precision). The LOQ values ranged from 0.020 µg/kg (L-PFOS) to 0.15 µg/kg (PFTeDA and PFTrDS), demonstrating good method sensitivity. Measurement uncertainty, expressed as relative expanded uncertainty (k = 2, ~95% confidence), was estimated from bias and precision and ranged from 16.6% (PFUnDA) to 62.4% (PFTeDA).

**Table 2.** Validation parameters for target PFAS analytes in eggs.

| Analyte | LOQ (µg/kg) | Recovery (%) | Relative Standard Deviation (%$RSD_R$) | Relative Expanded Uncertainty (k = 2) |
|---|---|---|---|---|
| Perfluoroalkyl carboxylic acids (PFCA) | | | | |
| PFHxA | 0.025 | 103.4 | 9.3 | 25.3 |
| PFHpA | 0.025 | 106.2 | 7.9 | 23.4 |
| PFOA | 0.05 | 104.3 | 11.0 | 28.4 |
| PFNA | 0.025 | 106.4 | 5.6 | 22.7 |
| PFDA | 0.025 | 114.1 | 14.5 | 22.7 |
| PFUnDA | 0.05 | 99.9 | 10.4 | 16.6 |
| PFDoDA | 0.025 | 105.7 | 21.7 | 31.9 |
| PFTrDA | 0.10 | 97.9 | 16.6 | 46.8 |
| PFTeDA | 0.15 | 109.7 | 15.7 | 62.4 |
| Perfluoroalkyl sulfonic acids (PFSA) | | | | |
| PFBS | 0.025 | 104.2 | 8.9 | 26.0 |
| PFPeS | 0.025 | 99.1 | 8.0 | 25.8 |
| PFHxS | 0.025 | 109.9 | 9.2 | 25.1 |
| PFHpS | 0.025 | 94.1 | 8.3 | 32.9 |
| L-PFOS | 0.020 | 111.7 | 9.9 | 27.8 |
| br-PFOS | 0.10 | 104.0 | 17.4 | 30.5 |
| PFNS | 0.05 | 86.0 | 23.2 | 31.4 |

**Table 2.** *Cont.*

| Analyte | LOQ (µg/kg) | Recovery (%) | Relative Standard Deviation (%$RSD_R$) | Relative Expanded Uncertainty (k = 2) |
|---|---|---|---|---|
| PFDS | 0.025 | 126.2 | 17.9 | 31.4 |
| PFUnDS | 0.10 | 76.7 | 16.9 | 46.8 |
| PFDoDS | 0.10 | 102.3 | 18.1 | 31.7 |
| PFTrDS | 0.15 | 72.0 | 12.3 | 48.8 |
| Perfluoroalkane sulfonamides | | | | |
| FOSA | 0.025 | 104.3 | 9.1 | 28.5 |
| PFAS Substitutes | | | | |
| DONA | 0.05 | 93.9 | 18.7 | 36.9 |
| HFPO-DA (Gen X) | 0.025 | 104.3 | 11.8 | 32.8 |
| 9Cl-PF3ONS | 0.05 | 96.8 | 15.8 | 30.7 |
| 11Cl-PF3OUdS | 0.025 | 128.0 | 17.2 | 17.6 |
| Capstone A | 0.025 | 101.4 | 24.8 | 48.7 |
| Capstone B | 0.05 | 93.6 | 23.9 | 45.5 |
| Fluoroteleomer sulfonates (FTS) | | | | |
| 4:2 FTS | 0.05 | 88.8 | 16.6 | 25.4 |
| 6:2 FTS | 0.05 | 88.9 | 16.6 | 24.7 |
| 8:2 FTS | 0.05 | 88.8 | 18.7 | 32.1 |

Consistent with other studies, only one MS/MS transition was found for PFBA and PFPeA [24,37,38]. The LOQ values obtained in this study were lower than those reported by Lasters et al. [27]. However, when compared with the studies by Lasters et al. [39], Mikołajczyk et al. [40], and Nobile et al. [41], the LOQ values were comparable or varied slightly, whereas Aßhoff et al. [42], reported lower LOQ values than those obtained in this study.

### *3.2. Occurrence of PFAS in Eggs*

In this study, the concentrations of PFAS compounds were determined in a total of 424 egg samples using a sensitive and accurate UHPLC-MS/MS method. The collected egg samples were categorized into four groups based on different hen rearing methods during egg production: cage farming, barn (stable) farming, organic farming, and home-produced eggs.

Of the 30 PFAS compounds analyzed, 13 were quantified. These included long-chain perfluoroalkyl carboxylic acids (LC-PFCA; >C7): PFOA, PFNA, PFDA, PFUnDA, PFDoDA, PFTrDA, PFTeDA, and PFUnDA. Among the long-chain perfluoroalkane sulfonic acids (LC-PFSA; >C6), PFHxS, PFHpS, linear PFOS (L-PFOS), and branched PFOS (br-PFOS) were quantified. Additionally, the compounds perfluoroalkane sulfonamide capstone B and fluorotelomer sulfonate 8:2 FTS were also quantified. The frequency of detection of PFAS compounds above the LOQ value is shown in Figure 1. Short-chain PFAS compounds (e.g., PFBS, or PFHxA) were not detected. The highest number of PFAS compounds (11) was quantified in home-produced eggs, followed by 9 in barn eggs, 5 in cage eggs, and only 3 in organic eggs. The remaining 18 PFAS compounds were not quantified above the LOQ.

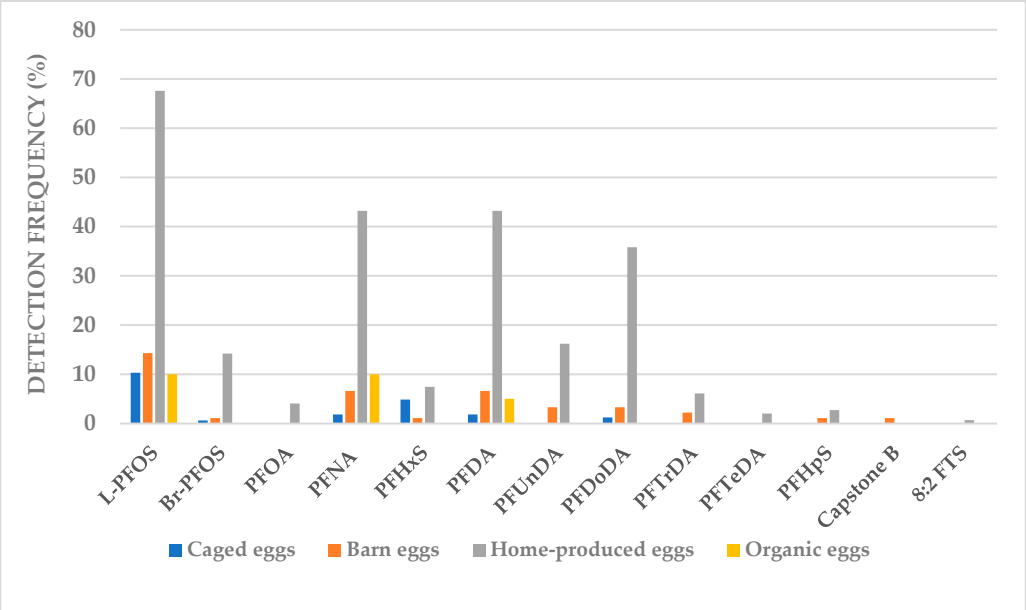



**Figure 1.** Detection frequency (%) of quantified PFAS compounds in cage, barn, organic and home-produced eggs in Croatia.

PFOS was quantified most frequently, ranging from 10% to 67.6%, while PFNA and PFDA were detected in 1.82% to 43.2% of samples. In home-produced eggs, the highest detection frequencies were observed for PFOS (67.6%), PFNA (43.2%), PFDA (43.2%), and PFDoDA (35.8%). Significantly lower detection frequencies for these four compounds—14.3%, 6.6%, 6.6%, and 3.3%, respectively—were found in barn eggs. In cage-farmed eggs, the highest detection frequencies were for PFOS (10.3%) and PFHxS (4.85%). The three compounds quantified in organic eggs—PFOS, PFNA, and PFDA—had detection frequencies of 10%, 10%, and 5%, respectively. PFOA was detected only in home-produced eggs, at a frequency of 4.05%.

Table 3 presents the concentrations of quantified PFAS compounds, including the sum of PFOS isomers and the sum of the four main PFAS compounds (∑4PFAS), in eggs from four different farming systems, as well as the total sum of compound concentrations for all eggs combined. The L-PFOS isomer was detected at the highest concentration of 3.85 μg/kg in barn eggs. The combined concentration of the two ∑PFOS isomers showed the highest levels among all quantified compounds across all four egg groups, with lower bound (LB) mean concentrations ranging from 0.070 to 0.23 μg/kg. Additionally, the highest mean LB and upper bound (UB) concentrations of Br-PFOS and L-PFOS isomers—i.e., ∑PFOS (0.23 and 0.32 μg/kg), PFNA (0.027 and 0.041 μg/kg), and PFHxS (0.003 and 0.026 μg/kg)—as well as the sum of these four compounds (∑4PFAS: 0.26 and 0.44 μg/kg) were observed in home-produced eggs. Home-produced eggs exhibited three to six times higher concentrations of ∑PFOS, PFNA, and PFHxS compared to cage and barn eggs, and more than 30 times higher compared to organic eggs. Statistical analysis revealed that home-produced eggs (LB and UB values) had significantly higher concentrations of L-PFOS, Br-PFOS, ∑PFOS, PFNA, and ∑4PFAS compared to the other egg types ($p$ = 0.0001, all).

**Table 3.** Concentrations of quantified PFAS compounds, including the sum of PFOS isomers, the sum of the four main PFAS compounds (∑4PFAS), and the total sum across all eggs, in eggs from four different farming systems.

| Compound | Statistics | Cage Eggs (µg/kg) | | Barn Eggs (µg/kg) | | Organic Eggs (µg/kg) | | Home-Produced (µg/kg) | | Total (µg/kg) | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | LB | UB | LB | UB | LB | UB | LB | UB | LB | UB |
| Four main PFAS | | | | | | | | | | | |
| Br-PFOS [a] | Mean ± SD | 0.20 | | 0.54 | | <LOQ | | 0.031 ± 0.11 | 0.12 ± 0.082 | 0.013 ± 0.069 | 0.11 ± 0.054 |
| | Range | | | | | | | 0.0–1.00 | 0.10–1.00 | 0.0–1.00 | 0.10–1.00 |
| L-PFOS [a] | Mean ± SD | 0.010 ± 0.061 | 0.028 ± 0.058 | 0.059 ± 0.408 | 0.076 ± 0.41 | 0.007 ± 0.022 | 0.025 ± 0.016 | 0.20 ± 0.38 | 0.21 ± 0.37 | 0.086 ± 0.31 | 0.10 ± 0.30 |
| | Range | 0.0–0.705 | 0.020–0.705 | 0.0–3.85 | 0.02–3.85 | 0.0–0.084 | 0.020–0.084 | 0.0–2.64 | 0.020–2.64 | 0.0–3.85 | 0.020–3.85 |
| ∑PFOS [a] | Mean ± SD | 0.011 ± 0.075 | 0.13 ± 0.066 | 0.065 ± 0.46 | 0.18 ± 0.45 | 0.007 ± 0.022 | 0.13 ± 0.016 | 0.23 ± 0.47 | 0.32 ± 0.44 | 0.10 ± 0.37 | 0.21 ± 0.35 |
| | Range | 0.0–0.91 | 0.12–0.91 | 0.0–4.39 | 0.12–4.39 | 0.0–0.084 | 0.12–0.18 | 0.0–3.64 | 0.12–3.64 | 0.0–4.39 | 0.12–4.39 |
| PFOA | Mean ± SD | <LOQ | | <LOQ | | <LOQ | | 0.003 ± 0.019 | 0.051 ± 0.011 | 0.003 ± 0.019 | 0.051 ± 0.011 |
| | Range | | | | | | | 0.0–0.17 | 0.050–0.17 | 0.0–0.17 | 0.050–0.17 |
| PFNA [a] | Mean ± SD | 0.001 ± 0.006 | 0.025 ± 0.003 | 0.004 ± 0.020 | 0.028 ± 0.015 | 0.003 ± 0.009 | 0.025 ± 0.002 | 0.027 ± 0.045 | 0.041 ± 0.037 | 0.011 ± 0.031 | 0.031 ± 0.024 |
| | Range | 0.0–0.064 | 0.025–0.064 | 0.0–0.15 | 0.025–0.15 | 0.0–0.032 | 0.025–0.032 | 0.0–0.34 | 0.025–0.34 | 0.0–0.34 | 0.025–0.34 |
| PFHxS [a] | Mean ± SD | 0.002 ± 0.010 | 0.026 ± 0.005 | 0.050 | <LOQ | <LOQ | | 0.003 ± 0.011 | 0.026 ± 0.005 | 0.002 ± 0.01 | 0.026 ± 0.005 |
| | Range | 0.0–0.060 | 0.025–0.060 | | | | | 0.0–0.065 | 0.025–0.065 | 0.0–0.065 | 0.025–0.065 |
| ∑4PFAS [a] | Mean ± SD | 0.014 ± 0.082 | 0.23 ± 0.069 | 0.070 ± 0.48 | 0.28 ± 0.47 | 0.010 ± 0.031 | 0.23 ± 0.017 | 0.26 ± 0.53 | 0.44 ± 0.48 | 0.12 ± 0.41 | 0.32 ± 0.37 |
| | Range | 0.0–0.97 | 0.22–1.04 | 0.0–4.54 | 0.22–4.61 | 0.0–0.11 | 0.22–0.29 | 0.0–3.93 | 0.22–3.93 | 0.0–4.54 | 0.22–4.54 |
| Perfluoroalkyl carboxylic acids (PFCA) | | | | | | | | | | | |
| PFDA [a] | Mean ± SD | 0.001 ± 0.008 | 0.027 ± 0.006 | 0.004 ± 0.017 | 0.027 ± 0.012 | 0.028 | | 0.027 ± 0.049 | 0.042 ± 0.042 | 0.011 ± 0.033 | 0.032 ± 0.027 |
| | Range | 0.0–0.090 | 0.025–0.090 | 0.0–0.12 | 0.025–0.12 | | | 0.0- 0.38 | 0.025–0.38 | 0.0–0.38 | 0.025–0.38 |
| PFUnDA [a] | Mean ± SD | <LOQ | | 0.004 ± 0.028 | 0.053 ± 0.021 | <LOQ | | 0.015 ± 0.040 | 0.057 ± 0.026 | 0.006 ± 0.028 | 0.053 ± 0.018 |
| | Range | | | 0.0–0.25 | 0.05–0.25 | | | 0.0–0.26 | 0.05–0.26 | 0.0–0.26 | 0.050–0.26 |
| PFDoDA [a] | Mean ± SD | 0.001 ± 0.006 | 0.026 ± 0.003 | 0.003 ± 0.015 | 0.027 ± 0.010 | <LOQ | | 0.022 ± 0.040 | 0.038 ± 0.033 | 0.008 ± 0.027 | 0.03 ± 0.021 |
| | Range | 0.0–0.067 | 0.025–0.067 | 0.0–0.11 | 0.025–0.11 | | | 0.0–0.23 | 0.025–0.23 | 0.0–0.23 | 0.025–0.23 |
| PFTrDA [b] | Mean ± SD | <LOQ | | 0.004 ± 0.031 | 0.10 ± 0.018 | <LOQ | | 0.010 ± 0.045 | 0.104 ± 0.026 | 0.005 ± 0.031 | 0.10 ± 0.018 |
| | Range | | | 0.0–0.28 | 0.10–0.28 | | | 0.0–0.34 | 0.10–0.34 | 0.0–0.34 | 0.10–0.34 |
| PFTeDA | Mean ± SD | <LOQ | | <LOQ | | <LOQ | | 0.003 ± 0.023 | 0.15 ± 0.002 | 0.001 ± 0.014 | 0.15 ± 0.001 |
| | Range | | | | | | | 0.0–0.17 | 0.15–0.17 | 0.0–0.17 | 0.15–0.17 |
| Perfluoroalkyl sulfonic acids (PFSA) | | | | | | | | | | | |
| PFHpS | Mean ± SD | <LOQ | | 0.030 | | <LOQ | | 0.001 ± 0.005 | 0.025 ± 0.001 | 0.0004 ± 0.003 | 0.025 ± 0.001 |
| | Range | | | | | | | 0.0–0.038 | 0.025–0.038 | 0.0–0.038 | 0.025–0.038 |
| Perfluoroalkane sulfonamides | | | | | | | | | | | |
| Capstone B | Mean ± SD | <LOQ | | 0.068 | | <LOQ | | <LOQ | | 0.068 | |
| | Range | | | | | | | | | | |
| Fluorotelomer sulfonates (FTS) | | | | | | | | | | | |
| 8:2 FTS | Mean ± SD | <LOQ | | <LOQ | | <LOQ | | 0.060 | | 0.060 | |
| | Range | | | | | | | | | | |

[a] Significant differences in PFAS levels between the four different farming systems of egg production: [a] $p = 0.0001$; [b] $p = 0.0071$.

PFOA was detected only in home-produced eggs, with a mean LB concentration of 0.0034 μg/kg. Among long-chain perfluorocarboxylic acids (LC-PFCA), PFDA and PFDoDA were quantified at the highest LB mean concentrations of 0.027 μg/kg and 0.022 μg/kg, respectively, in home-produced eggs. Statistical analysis further demonstrated that home-produced eggs had significantly higher concentrations of PFDA and PFDoDA ($p = 0.0001$ for both), as well as PFTrDA ($p = 0.0071$) and PFUnDA ($p = 0.0001$), compared to cage and barn eggs.

Among the LC-PFAS, the compound PFHpS was quantified in one barn egg (0.030 μg/kg) and detected at a mean concentration of 0.001 μg/kg in home-produced eggs. The compounds capstone B and 8:2 FTS were each quantified in one sample of eggs from either a stable or home-produced eggs, at concentrations of 0.068 μg/kg and 0.060 μg/kg, respectively.

In this study, primarily long-chain PFAS compounds were detected in eggs, consistent with previous research on these compounds in eggs [23,24,32,41,42]. Bioaccumulation of PFAS compounds in food chains is generally thought to increase with carbon chain length [2,26]. When comparing compounds of the same carbon chain length, studies have shown that PFSAs accumulate to a greater extent than PFCAs [2,43]. Short-chain PFASs have shorter half-lives and, consequently, lower bioaccumulation potential [43,44].

The significantly lower PFAS contamination observed in eggs from organic hens in this study is likely due to their diet of commercial feed, which is considerably less contaminated with PFAS compounds [23,24].

### *3.3. Comparison of PFAS Concentrations in Eggs from Other Studies*

Studies investigating PFAS contamination in the terrestrial food chain have shown that eggs contain the highest concentrations of PFAS among animal products and exhibit statistically significantly higher maximum concentrations for a greater number of detected PFAS compounds compared to other foods [2,4,45]. This is attributed to the fact that PFAS compounds bind more readily to proteins than to fats, resulting in higher concentrations in animal-derived foods such as tissues and eggs [2,39,46,47].

Investigations of PFAS contamination in eggs produced under different laying hen housing conditions in European countries are summarized in Table 4. Similar to this study, a recent investigation conducted in Italy analyzed 18 PFAS compounds across three groups of eggs (organic, caged, and barn). Seven long-chain PFAS compounds (PFHpA, PFNA, PFDA, PFDoDA, PFHxS, PFOS, and PFDoDS) were quantified at concentrations above the LOD [24]. Notably, no short-chain PFAS compounds were detected. Six PFAS compounds (PFHpA, PFNA, PFDA, PFDoDA, PFOS, PFDoDS) were found in eggs from organic farming, with the highest mean concentration of 0.026 μg/kg observed for PFDA in caged eggs. PFOS concentrations were similar in organic and barn eggs, whereas PFOS was below the LOD in caged eggs. The maximum measured concentration of PFOS was 0.042 μg/kg in barn eggs. Compared to eggs from the same farming types in this study, PFOS concentrations in cage and barn eggs were lower, while PFNA and PFHxS concentrations were higher. Italian organic eggs exhibited significantly higher concentrations of PFOS and PFNA than organic eggs in this study.

**Table 4.** PFAS concentrations (µg/kg) in eggs from European countries.

| Country | Egg Production Type | Statistics | N | PFOS | PFOA | PFNA | PFHxS | ∑4PFAS | PFDA | PFUnDA | PFDoDA | PFHpS | Reference |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Greece | Domestic eggs (yolk) | Median<br>Range<br>n > LOQ (%) | 45 | 1.10<br><0.5–8.9<br>69 | 0.50<br><0.5–0.5<br>2 | 0.80<br><0.5–1.0<br>20 | <0.5<br><0.5 | | 0.9<br><0.5–8.0<br>36 | | | <0.5<br><0.5 | [23] |
| The Nether-lands | Domestic eggs (yolk) | Median<br>Range<br>n > LOQ (%) | 73 | 3.50<br><0.5–24.8<br>81 | 1.10<br><0.5–2.7<br>27 | 0.90<br><0.5–2.0<br>18 | 1.1<br><0.5–5.2<br>7 | | 0.9<br><0.5–3.0<br>32 | | | <0.5<br><0.5 | [23] |
| Italy | Back-yard eggs | Mean [a]<br>Range<br>n > LOD | 78 | 0.66/0.73<br>0.25–3.47<br>51 | 0.01/0.14<br>0.62<br>16 | 0.03/0.18<br>0.25–1.20<br>29 | 0.03/0.15<br>0.25–0.50<br>19 | 0.73/1.20<br>0.25–4.64<br>58 | | | | | [48] |
| Italy | Organic eggs | Mean<br>Range<br>n > LOD | 57 | 0.016<br>0.007–0.025<br>4 | | 0.018<br>0.007–0.037<br>3 | <0.005<br><br>0 | | 0.021<br>0.014–0.030<br>3 | | 0.017<br>0.013–0.022<br>5 | | [24] |
| Italy | Caged hen eggs | Mean<br>Range<br>n > LOD | 27 | <0.006<br><br>0 | | 0.008<br>0.007–0.010<br>4 | 0.006<br><br>1 | | 0.026<br>0.018–0.032<br>4 | | 0.016<br>0.009–0.022<br>4 | | [24] |
| Italy | Barn eggs | Mean<br>Range<br>n > LOD | 27 | 0.015<br>0.007–0.042<br>5 | | 0.015<br><br>1 | 0.02<br>0.006–0.034<br>2 | | 0.014<br><br>1 | | <0.006<br><br>0 | | [24] |
| Italy | Commercial eggs | Mean<br>Range<br>n > LOQ | 65 | 0.05<br>0.05–0.5<br>2 | <LOQ | <LOQ | | | | <LOQ | | | [41] |
| Poland | Free range | Mean [a]<br>Median<br>Range<br>n > LOQ (%) | 15 | 0.11<br>0.09<br>0.01–0.25<br>30 | <LOQ | 0.02<br>0.02<br>0.01–0.03<br>27 | <LOQ | 0.04/0.11 | <LOQ | 0.02<br>0.02<br><LOQ-0.02<br>13 | <LOQ | <LOQ | [32] |
| | Organic | Mean<br>Median<br>Range<br>n > LOQ (%) | 15 | 0.13<br>0.02<br><LOQ-0.74<br>60 | 0.046<br>0.046<br><LOQ-0.46 | 0.04<br>0.03<br>0.02–0.09<br>27 | <LOQ | 0.10/0.17 [a] | 0.06<br>0.06<br><LOQ-0.06 | 0.04<br>0.03<br><LOQ-0.08<br>27 | 0.06<br>0.06<br><LOQ-0.06<br>7.00 | <LOQ | [32] |
| Poland | Baterry cage | Mean [a]<br>Median<br>Range<br>n > LOQ (%) | 15 | <LOQ | <LOQ | <LOQ | <LOQ | 0.0/0.08 | <LOQ | 0.02<br>0.02<br><LOQ-0.02<br>7 | 0.06<br>0.06<br><LOQ-0.06<br>7 | <LOQ | [32] |

**Table 4.** *Cont.*

| Country | Egg Production Type | Statistics | N | PFOS | PFOA | PFNA | PFHxS | ∑4PFAS | PFDA | PFUnDA | PFDoDA | PFHpS | Reference |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Poland | Market | Mean<br>Range<br>n > LOQ | 5 | 2.40<br>0.064–5.7<br>5 | 0.13<br>0.10–0.18<br>5 | 0.08<br><LOQ-0.17<br>3 | 0.03<br><LOQ-0.14<br>2 | | <LOQ-0.11<br>0 | | | | [45] |
| Poland | Free range and cage | Mean [b]<br>Median<br>Range<br>n > LOQ (%) | 25 | 0.081<br>0.081<br><LOQ-0.095 | <LOQ | 0.031<br>0.031<br><LOQ-0.031 | <LOQ | 0.096<br>0.096<br>0.094–0.097 | <LOQ | 0.03<br>0.03<br><LOQ-0.03 | | <LOQ | [40] |
| Germany | Market | Mean<br>Median<br>Range<br>n > LOQ | 56 | 0.201<br>0.14<br>0.02–0.772<br>40 | 0.049<br>0.03<br>0.02–0.194<br>25 | 0.031<br>0.03<br>0.02–0.057<br>25 | 0.038<br>0.03<br>0.023–0.06<br>9 | | 0.04<br>0.04<br>0.022–0.083<br>24 | | 0.06<br>0.04<br>0.025–0.113<br>9 | | [42] |
| Croatia | Cage | Mean [a] | 165 | 0.11 | <LOQ | 0.001 | 0.002 | 0.014 | 0.001 | <LOQ | 0.001 | <LOQ | This study |
| | | Range | | 0.0–0.91 | | 0.0–0.064 | 0.0–0.06 | 0.0–0.97 | 0.0–0.09 | | 0.0–0.067 | | |
| | | n > LOQ (%) | | 10.3 | 0 | 1.82 | 4.85 | | 1.82 | | 1.21 | | |
| Croatia | Barn | Mean [a] | 91 | 0.065 | <LOQ | 0.004 | 0.05 [c] | 0.07 | 0.004 | 0.004 | 0.003 | 0.03 [c] | This study |
| | | Range | | 0.0–4.39 | | 0.0–0.15 | | 0.0–4.54 | 0.0–0.12 | 0.0–0.25 | 0.0–0.11 | | |
| | | n > LOQ (%) | | 14.3 | 0 | 6.60 | 1.10 | | 6.60 | 3.30 | 3.30 | 1.10 | |
| Croatia | Organic | Mean [a] | 20 | 0.007 | <LOQ | 0.003 | <LOQ | 0.010 | 0.028 [c] | <LOQ | <LOQ | <LOQ | This study |
| | | Range | | 0.0–0.084 | | 0.0–0.032 | | 0.0–0.11 | | | | | |
| | | n > LOQ (%) | | 10.0 | 0 | 10.0 | 0 | | 5.0 | 0 | 0 | 0 | |
| Croatia | Home-produced | Mean [a] | 148 | 0.023 | 0.003 | 0.027 | 0.003 | 0.26 | 0.027 | 0.015 | 0.022 | 0.001 | This study |
| | | Range | | 0.0–3.64 | 0.0–0.17 | 0.0–0.34 | 0.0–0.065 | 0.0–3.93 | 0.0–0.38 | 0.0–0.26 | 0.0–0.23 | 0.0–0.038 | |
| | | n > LOQ (%) | | 67.6 | 4.05 | 43.2 | 7.43 | | 43.2 | 16.2 | 35.8 | 2.70 | |

[a] LB, lower bond/UB, upper bond; [b] LB, lower bond; [c] only one result.

Significantly higher concentrations of PFOS, PFNA, PFHxS, PFDA, and PFDoDA were found in commercial eggs from Poland [40,45] and Germany [42] compared to the levels measured in caged and barn organic eggs in this study. PFOA concentrations in these commercial eggs were reported as 0.13 μg/kg [45], 0.013 μg/kg [40], and 0.049 μg/kg [42], whereas PFOA was not detected in any of the three types of commercially produced eggs analyzed in this study. Additionally, the mean LB concentrations of PFOS, PFNA, ∑4PFAS, and PFUnDA in caged and barn eggs from this study were lower than those found in free-range and caged eggs from Poland [40]. Median concentrations of PFOS, PFOA, PFNA, and PFDA measured in the yolks of domestic eggs from Greece and the Netherlands were significantly higher than the median values determined in home-produced eggs in this study (0.077, 0.000, 0.000, and 0.000 μg/kg, respectively) [23]. In this research, maximum PFOS concentrations of 24.8 μg/kg were recorded in eggs from the Netherlands. PFOS and PFOA were detected in 69% and 2% of Greek eggs, respectively, which is comparable to the detection frequencies observed in this study. However, detection frequencies in eggs from the Netherlands were higher, at 81% for PFOS and 27% for PFOA. Conversely, PFNA was detected at lower frequencies—20% and 18%—in eggs from Greece and the Netherlands, respectively, compared to this study. However, these figures are not directly comparable, as they refer specifically to egg yolk, which tends to concentrate PFAS to a greater extent than the whole eggs.

In backyard eggs from Italy, PFOS is the most frequently detected compound, with a detection frequency of 65.4% [48], similar to that observed in home-produced eggs in this study. PFOA and PFHxS were detected at higher frequencies—20.5% and 24.4%, respectively—while PFNA was detected less frequently (37.2%) compared to this study. However, concentrations of PFOS (0.66 μg/kg), PFOA (0.01 μg/kg), and ∑4PFAS (0.73 μg/kg) in the Italian backyard eggs were approximately three times higher, and PFHxS concentrations (0.03 μg/kg) were about ten times higher, whereas PFNA concentrations (0.03 μg/kg) were similar to those found in this study.

In Polish eggs from organic farming [32], higher mean concentrations of PFOS, PFNA, and ∑4PFAS were reported compared to organic eggs in this study. Additionally, PFOA (0.046 μg/kg), PFUnDA (0.04 μg/kg), and PFDoDA (0.06 μg/kg) were quantified in the Polish samples, in contrast to the organic eggs analyzed in this study.

The total mean LB and UB concentrations for ∑PFOS obtained in this study were 0.103 μg/kg and 0.207 μg/kg, respectively, which are lower than the LB (0.27 μg/kg) and UB (0.35 μg/kg) values reported by EFSA in their consolidated European report [10]. The LB values for PFOA, PFNA, and PFHxS in this study are higher than those published by EFSA (0.106, 0.0001, and 0.0002 μg/kg, respectively), while the UB values are lower than the EFSA values (0.21, 0.098, and 0.06 μg/kg, respectively). It is necessary to emphasize that EFSA stated in the same report that the LB values are probably more realistic than the UB values [10].

The results of this study are consistent with previous research demonstrating a lower incidence of egg contamination in commercial eggs produced by layers raised in cages and barns. These eggs exhibited lower concentrations of ∑4PFAS and other LC-PFAS compounds compared to home-produced eggs [23,24,32,48]. Home or backyard eggs are common in rural areas and smaller towns, where a considerable number of people keep chickens in their yards and produce eggs for home consumption. These chickens are typically kept outdoors, allowed to roam freely within the yard or a larger area, and are therefore exposed to various environmental pollutants, including PFAS [23,48]. Due to their stability and resistance to various forms of degradation, PFAS compounds persist in the environment for extended periods. Free-range laying hens are exposed to these compounds to a much greater extent than those raised under controlled housing and

feeding conditions. Furthermore, the conditions under which laying hens are raised in free-range systems—such as the shape and size of the space, soil characteristics, vegetation, and biological factors—significantly influence their exposure to PFAS compounds [27]. Raising laying hens in household backyards involves the unrestricted use of soil as a substrate for nutrition, with the hens feeding on household food scraps, insects and plants. Each of these is a potential source of PFAS exposure for laying hens, with soil identified as one of the primary sources [27,49,50]. Also, recent studies have demonstrated the transfer of PFAS from water and feed to eggs [44,51]. Therefore, current research primarily focuses on investigating soil contamination by PFAS compounds, with an even greater emphasis on surface water and drinking water, as these represent the most significant sources of animal and human exposure [7,52–54].

### 3.4. Weekly Intake Estimation

Among a total of 424 egg samples, nine eggs (eight home-produced and one barn egg) had PFOS concentrations above the ML value of 1 μg/kg. In contrast, the concentrations of the other three PFAS compounds (PFOA, PFNA, and PFHxS) were below the permitted ML values set by Regulation (EU) 2023/915 [14]. Therefore, the overall incidence of egg contamination above the ML for PFOS was 2.12%. For four home-produced egg samples and one barn egg, the sum of ∑4PFAS exceeded the permitted value of 1.7 μg/kg, resulting in a total incidence of only 1.18%.

Weekly intakes of the sum of ∑4PFAS through egg consumption were calculated for six different age groups of the Croatian population. These calculations considered total intake from all eggs, as well as from caged eggs and home-produced eggs, representing the lowest and highest values of the sum of these four compounds, respectively. Exposure levels were estimated using both LB and UB values, representing a "worst-case" scenario. These were compared with the TWI limit of 4.4 ng/kg bw per week and expressed as %TWI [10]. The results are presented in Table 5. These findings represent the first assessment of the Croatian population's exposure to PFAS compounds through food consumption.

**Table 5.** Estimated weekly intake calculated for LB and UB values of ∑4PFAS, with risk characterization relative to TWI for different age groups of the Croatian population.

| Age Group | Production Type | Weekly Intake (ng/kg bw) | | | |
|---|---|---|---|---|---|
| | | LB | % TWI | UB | % TWI |
| Infants | Caged | 0.18 | 4.14 | 2.93 | 66.6 |
| | Home-produced | 3.35 | 76.2 | 5.61 | 127.4 |
| | Total | 1.50 | 34.2 | 4.01 | 91.2 |
| Toddlers | Caged | 0.17 | 3.80 | 2.67 | 60.7 |
| | Home-produced | 3.06 | 69.5 | 5.11 | 116.2 |
| | Total | 1.37 | 31.2 | 3.66 | 83.2 |
| Children | Caged | 0.10 | 2.32 | 1.64 | 37.3 |
| | Home-produced | 1.88 | 42.7 | 3.14 | 71.4 |
| | Total | 0.84 | 19.1 | 2.25 | 51.1 |
| Adolescents | Caged | 0.055 | 1.25 | 0.89 | 20.1 |
| | Home-produced | 0.89 | 30.3 | 1.69 | 38.5 |
| | Total | 0.45 | 10.3 | 1.21 | 27.6 |
| Adults | Caged | 0.048 | 1.09 | 0.77 | 17.6 |
| | Home-produced | 0.78 | 17.7 | 1.48 | 33.6 |
| | Total | 0.39 | 9.01 | 1.06 | 24.1 |
| Eldery | Caged | 0.042 | 1.00 | 0.68 | 15.4 |
| | Home-produced | 0.68 | 15.5 | 1.29 | 29.4 |
| | Total | 0.35 | 7.90 | 0.93 | 21.0 |

Despite the fact that most samples fall below the ML set by EU regulation, the results indicate that egg consumption significantly contributes to the exposure of the Croatian population to ∑4PFAS, in particular, given that the exposure estimates were derived based on average egg consumption and not on consumption by high consumers. The lowest weekly intakes (LB and UB) for ∑4PFAS were calculated for caged eggs, while the highest were for home-produced eggs across all age groups. Exposure from home-produced eggs was more than 15 times higher than from caged eggs. For children up to 10 years of age—including infants, toddlers, and children—exposure relative to the TWI was determined as 76.2%, 69.5%, and 42.7% for LB values, and 127.4%, 116.2%, and 71.4% for UB values, respectively. Lower risk was observed for adolescents, adults, and elderly, with weekly LB intakes calculated at 0.89, 0.78 and 0.68 ng/kg bw, respectively.

Considering the overall Croatian population, the most representative exposure estimates are the mean values based on the total sum of ∑4PFAS for all eggs. The weekly LB intake was determined to be 1.5 ng/kg bw for infants, 1.37 ng/kg bw for toddlers, 0.84 ng/kg bw for children, 0.45 ng/kg bw for adolescents, 0.39 ng/kg bw for adults and 0.35 ng/kg bw for eldery. The highest contributions to TWI, for both LB and UB values, were observed in infants (34.2% and 91.2%), toddlers (31.2% and 83.2%), and children (19.1% and 51.1%). The adult and eldery population's contribution to TWI was 9.01% and 7.90% for LB, and 24.1% and 21.0% for UB values.

Significant exposure to these compounds was also demonstrated by an Italian study conducted across different age groups, focusing on the consumption of eggs from backyard chickens in northern, central, and southern Italy [48]. The highest contamination levels were found in eggs from northern Italy, and the highest exposure among children, with %TWI for LB and UB being 36% and 55%, respectively. The mean weekly exposure in Italy was 1.29 ng/kg bw for children, 0.64 ng/kg bw for adolescents, and 0.47 ng/kg bw for adults, which was higher compared to the corresponding age groups in this study. Conversely, a recent Italian study reported lower exposures than those found here for eggs from various farming types [24]. Weekly intakes of ∑4PFAS and %TWI, calculated based on UB values, were 0.76 ng/kg bw and 17.3% for children, 0.40 ng/kg bw and 9% for adolescents, and 0.31 ng/kg bw and 7% for adults. A parallel study from Italy, examining eggs from Northern Italian markets, reported a weekly intake value of 0.23 ng/kg bw for adults, corresponding to a 5.23% contribution to the TWI [41], which is lower than the values observed in this study.

## 4. Conclusions

The sensitive UHPLC-MS/MS method developed and applied in this study enabled the quantification of low concentrations of 30 PFAS compounds in eggs from laying hens raised in different housing systems. The results provide the first insight into the concentrations of these compounds in eggs and the overall contamination of animal-derived food in Croatia.

The findings reveal significant differences in PFAS concentrations depending on the production system, with the highest levels detected in home-produced eggs. The lowest concentrations were found in cage-produced eggs, which constitute the majority of eggs available on the Croatian market (67.3%). These results corroborate previous studies indicating that home-produced eggs are more contaminated with PFAS than cage or barn eggs. The mean concentrations observed in this study are generally lower than those reported in similar studies conducted in European countries and those published in the EFSA report.

Dietary exposure calculations, expressed as weekly intake of the sum of four PFAS (∑4PFAS) across six age groups in the Croatian population, indicate that children up to nine

years old are the most vulnerable to exposure. This vulnerability is especially pronounced for infants and toddlers consuming home-produced eggs, due to their low body weight. Given the significantly lower exposure associated with cage-produced eggs, consumers are advised to prefer eggs from this production system.

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